BUSINESS ANALYTICS | DESCRIPTIVE ANALYTICS COURSE

LAB REPORT

Power BI Practicals 7, 8 and 9

Dataset: RetailPulse Sales (Sales, Product and Store tables)

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Submitted to	Miss Monica
Course	Business Analytics (Descriptive Analytics)
Tool used	Microsoft Power BI Desktop
Dataset	RetailPulse Sales (2024 - 2025)

OPERATIONS COVERED

Practical	Topic	Power BI operations performed
7	Import and Transform Data	Get data (Excel), Power Query, data types, remove blank rows and duplicates, replace nulls, conditional column, Close & Apply
8	Model, Calculate and Publish	Relationships (Model view), measures, calculated column, bar / line / card visuals, Region slicer, cross-filtering, Publish
9	Time Intelligence and Advanced DAX	Date table, TOTALYTD, SAMEPERIODLASTYEAR, VAR/RETURN, CALCULATE with ALL, matrix, Performance Analyzer

Dataset Used: RetailPulse Sales

A clean, randomly generated retail dataset in the structure the lab manual expects (workbook: RetailPulse_Sales.xlsx).

Table	Rows	Columns	Role
Sales	1,002	OrderID, Sales Date, StoreID, ProductID, Quantity, Sales Amount	Fact table
Product	18	ProductID, Product Name, Product Category, Unit Price	Dimension
Store	10	StoreID, Store Name, Region, City	Dimension

Coverage: 1 Jan 2024 to 31 Dec 2025 (two full calendar years, as Practical 9 requires). **Regions:** North, South, East, West. **Categories:** Electronics, Sports, Apparel, Home & Kitchen, Grocery, Beauty. All amounts are in Indian rupees (Rs).

Data quality check before import: 0 blank cells, 0 duplicate rows, every OrderID unique. Because the dataset was already clean, the cleaning steps in Practical 7 were still applied but removed 0 rows.

Headline results (all verified from the data)

Measure	Value
Total Sales	Rs 5,088,178.10
Total Quantity	2,975
Average Order Value	Rs 5,078.02
Sales 2024	Rs 2,199,436.10
Sales 2025	Rs 2,888,742.00
YoY growth (2025 vs 2024)	31.3%

Practical 7: Import and Transform RetailPulse Data in Power BI

Related topic: Power BI Interface, Data Import and Transformation (Unit 3)

OBJECTIVE

Explore the Power BI Desktop interface, import the RetailPulse data from a workbook, and clean and shape it using Power Query.

MATERIALS REQUIRED

Power BI Desktop (free version), RetailPulse_Sales.xlsx workbook containing Sales, Product and Store tables.

STEP-BY-STEP PROCEDURE

1. Open Power BI Desktop and identify the Report, Table and Model views, the ribbon, and the Data, Visualizationsand Filters panes.
2. Select **Home > Get data > Excel workbook**, open RetailPulse_Sales.xlsx, tick the Sales, Product and Store tables in the Navigator, and click **Transform Data**.
3. In Power Query Editor set the data types: Sales Date as Date, Quantity as Whole Number, Sales Amount as Fixeddecimal number.
4. Remove blank rows and duplicate rows from the Sales table (Home > Remove Rows), replace null values inQuantity with 0 (Transform > Replace Values), and rename any unclear column headings.
5. Add a Conditional Column (Add Column > Conditional Column) named **Sale Size**: "Large" when Sales Amount is 1000 or more, otherwise "Standard".
6. Review the Applied Steps list, click **Close & Apply**, and save the file as RetailPulse_Analysis.pbix.

SAMPLE SOLUTION / RESULT

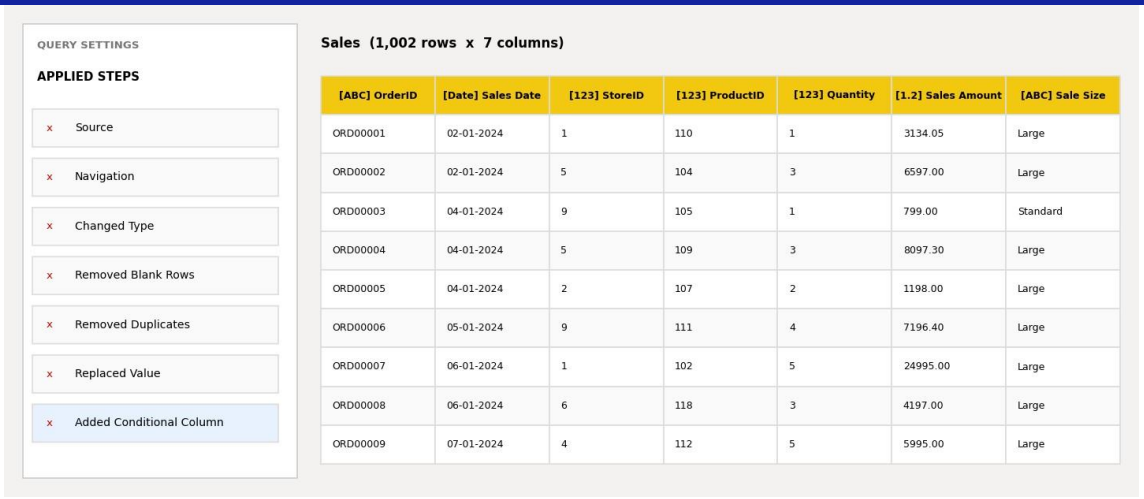


Figure 7.1: Power Query Editor - Applied Steps and the Sales table preview with the Sale Size column.

Check performed	Result
Blank rows removed	0 (dataset already clean)
Duplicate rows removed	0
Null values in Quantity replaced with 0	0 nulls found
Sale Size = Large (Sales Amount >= 1000)	913 rows
Sale Size = Standard (Sales Amount < 1000)	89 rows
Error values in any column	None

Conditional column (M code): `if [Sales Amount] >= 1000 then "Large" else "Standard"`

EXPECTED OUTCOME

All three tables load without errors; dates are recognised as dates and amounts as numbers; blank rows and duplicates are gone; the Sale Size column labels each row; the Applied Steps list records every change and re-runs on refresh.

VIVA-VOCE QUESTIONS AND ANSWERS

a) **Import mode vs DirectQuery?** Import copies the data into Power BI's in-memory model (fast, works offline, needs refresh). DirectQuery keeps data in the source and sends a query for every visual interaction (always current, but slower and with some feature limits).

b) **What are Applied Steps and why are they useful?** The ordered list of every transformation Power Query records. They can be edited, reordered or deleted and are replayed automatically each time the data refreshes.

c) **Which view checks relationships?** Model view.

d) **Why set data types first?** Visuals, sorting, aggregation and DAX depend on type. Text amounts cannot be summed and text dates cannot support time intelligence.

e) **Transform Data vs Close & Apply?** Transform Data opens Power Query Editor to shape the data. Close & Apply saves the transformations and loads the result into the model.

Practical 8: Model, Calculate and Publish a RetailPulse Sales Report

Related topic: Relationships, Basic DAX, Visualizations and Publishing Reports (Unit 3)

OBJECTIVE

Create relationships between the RetailPulse tables, write basic DAX measures and a calculated column, build interactive visuals, and publish the report to the Power BI service.

MATERIALS REQUIRED

Power BI Desktop, RetailPulse_Analysis.pbix from Practical 7, a Power BI service account for publishing.

STEP-BY-STEP PROCEDURE

1. Open **Model view** and create one-to-many relationships Product[ProductID] to Sales[ProductID] and Store[StoreID] to Sales[StoreID], single-direction filtering.
2. Create three measures: Total Sales, Total Quantity and Average Order Value (DAX below).
3. Create a calculated column in Sales named Sales per Unit.
4. Build a report page: a card for Total Sales, a bar chart of Total Sales by Product Category, a line chart of TotalSales by month, and a slicer for Region.
5. Click a bar and change the slicer to confirm all visuals cross-filter, then add clear titles and axis labels.
6. Select **Home > Publish**, choose a workspace, open the report in the Power BI service and share it with the instructor (subject to workspace permissions).

SAMPLE SOLUTION: DAX

```
Total Sales = SUM(Sales[Sales Amount])
Total Quantity = SUM(Sales[Quantity])
Average Order Value = DIVIDE([Total Sales], DISTINCTCOUNT(Sales[OrderID]))
Sales per Unit = DIVIDE(Sales[Sales Amount], Sales[Quantity]) (calculated column)
```

Model
and

Sales
many
and *

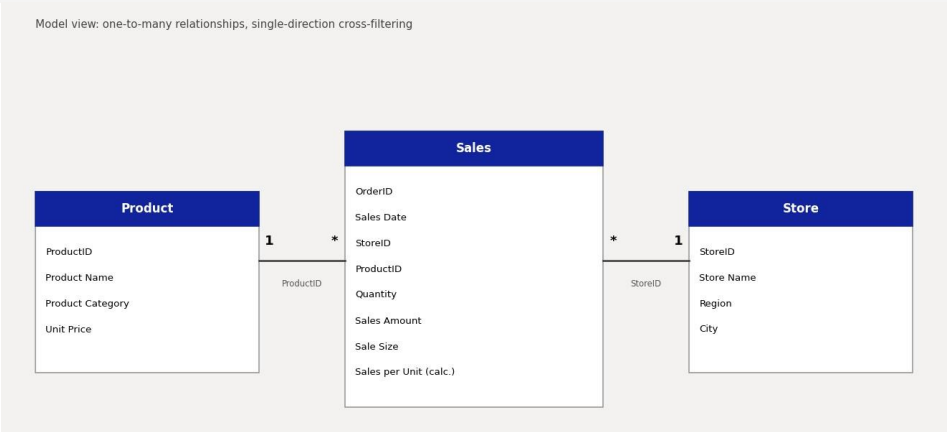


Figure 8.1: view - Product Store each connected to by a one-to- relationship (1 markers).

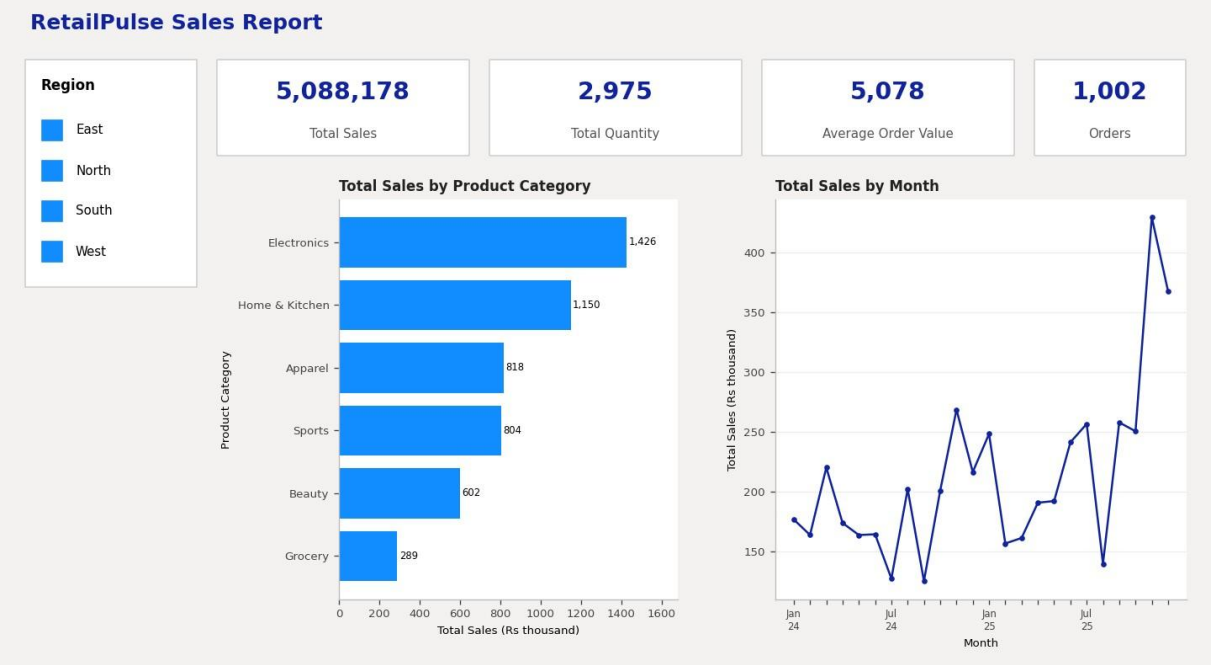


Figure 8.2: Report page - cards, bar chart by Product Category, line chart by month and the Region slicer.

Measure results (no slicer selection)

Measure	Value
Total Sales	5,088,178.10
Total Quantity	2,975
Average Order Value	5,078.02

Total Sales by Product Category (bar chart values)

Product Category	Total Sales	Share
Electronics	1,426,225.65	28.0%
Home & Kitchen	1,150,038.50	22.6%
Apparel	818,224.70	16.1%
Sports	803,560.60	15.8%
Beauty	601,614.15	11.8%
Grocery	288,514.50	5.7%
Total	**5,088,178.10**	**100.0%**

The bar chart values add up to the Total Sales card (5,088,178.10), confirming the relationships and measures work correctly.

EXPECTED OUTCOME

Relationship lines show 1 and * on the correct ends; the three measures and the calculated column return values; measures change when the Region slicer changes; the Total Sales card equals the sum of the bars. Publishing makes the report visible in the chosen workspace, and others can open it only if given access.

VIVA-VOCE QUESTIONS AND ANSWERS

a) Why one-to-many from dimension to fact? Each dimension row (one product or store) matches many fact rows. This keeps dimension data unique, filters flow cleanly into the fact table, and avoids ambiguity and duplication.

b) Measure vs calculated column? A measure is calculated at query time in the current filter context and is not stored per row. A calculated column is computed row by row at refresh and stored in the model.

c) Why DIVIDE instead of / ? DIVIDE handles division by zero (and blanks) safely, returning BLANK or an alternate result instead of an error.

d) Visual for a trend over time? A line chart, because it connects points in time order and shows direction and seasonality clearly.

e) What must be in place before others can view a published report? They need access to the workspace or the report (sharing permission) and a suitable licence, e.g. Pro or a Premium capacity.

Practical 9: Time Intelligence and Advanced DAX for RetailPulse

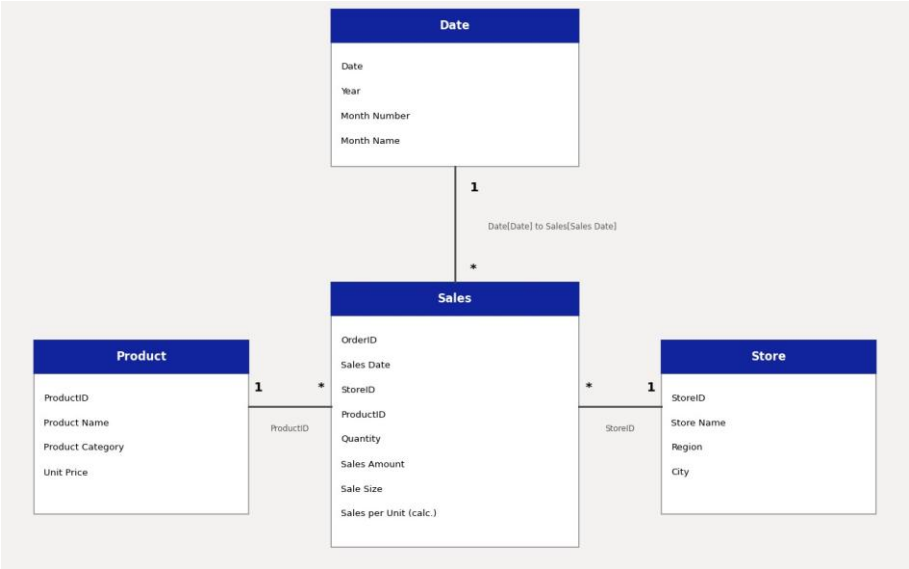
Related topic: Advanced Data Modeling, Time Intelligence, Complex DAX Functions and Performance Optimization (Unit 4)

OBJECTIVE

Create a date table, build time intelligence and context-modifying DAX measures, and use Performance Analyzer to find and reduce slow report elements.

MATERIALS REQUIRED

Power BI Desktop, RetailPulse_Analysis.pbix from Practical 8. Sales data covers two calendar years (2024 and 2025) so the year-on-year measures display values.



STEP-BY-STEP PROCEDURE

1. Create a Date table, add Year, Month Number and Month Name columns, mark it as a date table and relateDate[Date] to Sales[Sales Date].
2. Create the measures Sales YTD and Sales LY using TOTALYTD and SAMEPERIODLASTYEAR.
3. Create YoY Growth % using variables (VAR and RETURN).
4. Create Region Share % using CALCULATE with ALL to remove the Region filter from the denominator.
5. Build a matrix with Year and Month on rows and Total Sales, Sales YTD, Sales LY and YoY Growth % as values.
6. Open **View > Performance Analyzer**, record, refresh visuals and note each duration; then remove unused columns, reduce visuals on the page and record again to compare.

SAMPLE SOLUTION: DAX

```
Date = CALENDAR(MIN(Sales[Sales Date]), MAX(Sales[Sales Date]))
Year = YEAR('Date'[Date])
Month Number = MONTH('Date'[Date])
Month Name = FORMAT('Date'[Date], "MMMM")

Sales YTD = TOTALYTD([Total Sales], 'Date'[Date])
Sales LY = CALCULATE([Total Sales], SAMEPERIODLASTYEAR('Date'[Date]))
YoY Growth % = VAR Curr = [Total Sales]
  VAR Prev = [Sales LY]
  RETURN DIVIDE(Curr - Prev, Prev)
Region Share % = DIVIDE([Total Sales], CALCULATE([Total Sales], ALL(Store[Region])))
```

Figure 9.1: Model view with the Date table related to Sales[Sales Date] (one-to-many).

Matrix output: Total Sales, Sales YTD, Sales LY and YoY Growth %

Year	Month	Total Sales	Sales YTD	Sales LY	YoY Growth %
2024	January	176,417.95	176,417.95		
2024	February	163,485.70	339,903.65		
2024	March	219,851.10	559,754.75		
2024	April	173,519.30	733,274.05		
2024	May	163,406.75	896,680.80		
2024	June	164,049.50	1,060,730.30		
2024	July	126,950.20	1,187,680.50		
2024	August	201,736.30	1,389,416.80		
2024	September	124,878.55	1,514,295.35		
2024	October	200,672.65	1,714,968.00		
2024	November	268,335.85	1,983,303.85		
2024	December	216,132.25	2,199,436.10		
2025	January	247,997.50	247,997.50	176,417.95	40.6%
2025	February	156,356.75	404,354.25	163,485.70	-4.4%
2025	March	160,949.45	565,303.70	219,851.10	-26.8%
2025	April	190,454.80	755,758.50	173,519.30	9.8%
2025	May	191,855.15	947,613.65	163,406.75	17.4%
2025	June	240,997.95	1,188,611.60	164,049.50	46.9%
2025	July	256,362.40	1,444,974.00	126,950.20	101.9%
2025	August	139,417.60	1,584,391.60	201,736.30	-30.9%
2025	September	257,590.90	1,841,982.50	124,878.55	106.3%
2025	October	250,211.75	2,092,194.25	200,672.65	24.7%
2025	November	429,173.65	2,521,367.90	268,335.85	59.9%
2025	December	367,374.10	2,888,742.00	216,132.25	70.0%
2024 Total		**2,199,436.10**	**2,199,436.10**		
2025 Total		**2,888,742.00**	**2,888,742.00**	**2,199,436.10**	**31.3%**

2024 rows show a blank Sales LY and YoY Growth % because there is no 2023 data. Sales YTD restarts in January of each year.

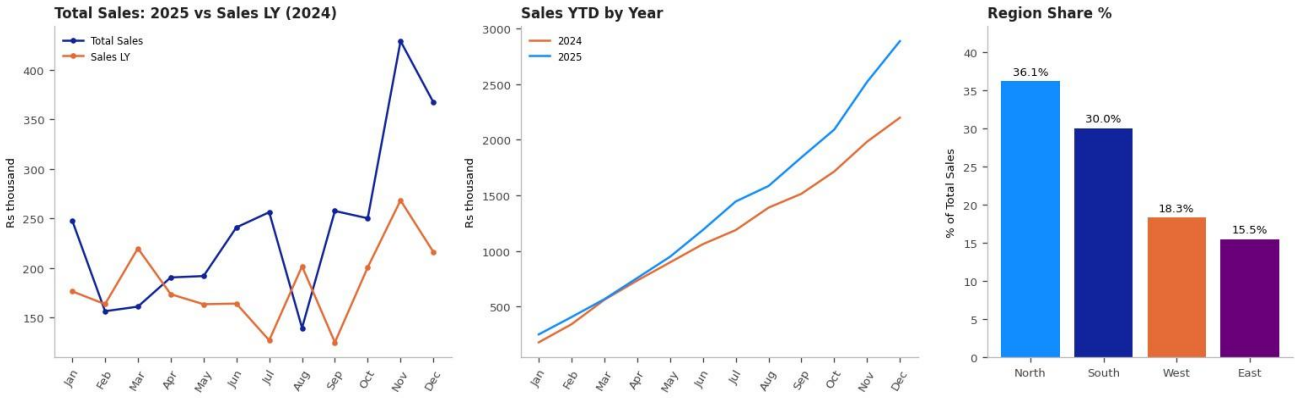


Figure 9.2: Sales LY comparison, Sales YTD by year and Region Share %.

Region Share % (CALCULATE with ALL)

Region	Total Sales	Region Share %
North	1,839,365.10	36.1%
South	1,527,906.35	30.0%
West	931,978.45	18.3%
East	788,928.20	15.5%
Total	**5,088,178.10**	**100.0%**

Performance Analyzer record

Timings depend on the machine, so they are recorded in Power BI Desktop itself. Fill in the durations (ms) shown by View > Performance Analyzer.

Visual	Before optimisation (ms)	After optimisation (ms)
Matrix		
Line chart		
Bar chart		
Card		

Changes made: removed unused columns, reduced the number of visuals on the page, then recorded again.

EXPECTED OUTCOME

Month rows reset YTD at the start of each year; first-year rows show blank Sales LY and YoY Growth %; Region Share % adds up to 100%; Performance Analyzer shows measurable timing per visual and a record of what was changed.

VIVA-VOCE QUESTIONS AND ANSWERS

a) **Why a separate continuous date table?** Time intelligence functions need an unbroken list of dates with no gaps to shift or accumulate periods correctly; a fact table's dates have gaps.

b) **What does CALCULATE do?** It evaluates an expression in a modified filter context: it adds, replaces or removes filters before computing.

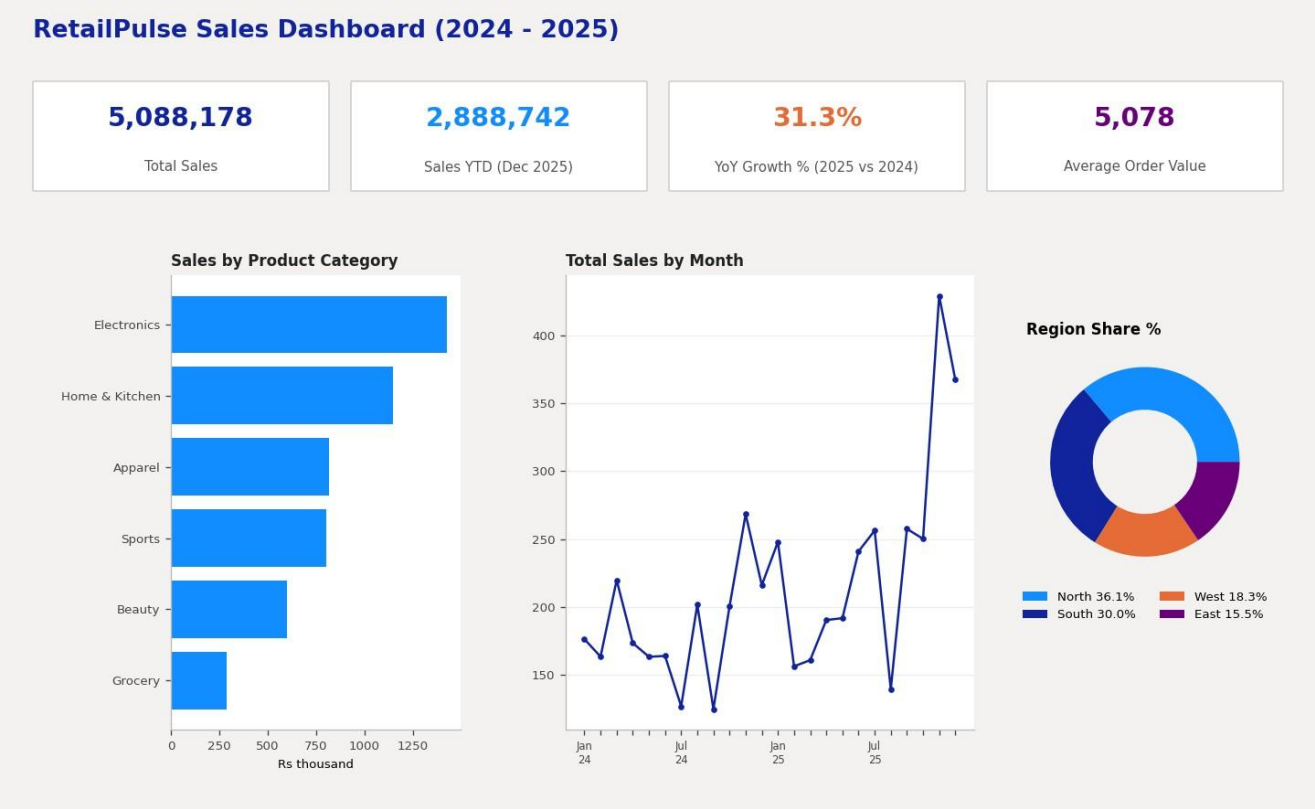
c) **Benefit of VAR?** The value is calculated once and reused, which improves readability and performance and makes debugging easier.

d) **What does ALL remove, and why in Region Share %?** ALL removes filters from the named column or table. It makes the denominator the grand total, so each region's share is measured against all regions.

e) **Two ways to improve performance?** Remove unused columns and tables; reduce the number of visuals per page; use efficient measures (variables); use a star schema.

Final Dashboard: RetailPulse Sales

Combines the Practical 8 and Practical 9 measures on one report page.



Key insights

1. Total sales were Rs 5,088,178; 2025 grew 31.3% over 2024.
2. Electronics is the top category (Rs 1,426,226); Grocery is the smallest (Rs 288,514).
3. North region contributes the largest share (36.1%), followed by South (30.0%).
4. Sales peak in the last quarter of each year (festive season), with the highest month in late 2025.